

Building Detection from Satellite Imagery using YOLOv8

Problem

In this project, I built a model that can detect buildings from satellite images and count how many buildings are present in an image. This type of system can be useful for city planning, mapping, disaster management, and monitoring urban development. The main goal was to train a model that can automatically identify buildings from aerial images.

Data

For this project, I used the SpaceNet Building Detection dataset from Kaggle. The dataset contains satellite images along with label files that show the location of buildings using bounding boxes. There were 1148 images in total. The dataset was divided into training, validation, and testing sets using an 80:10:10 ratio.

Model

I used YOLOv8s for building detection. YOLO (You Only Look Once) is a popular object detection model that can identify objects and their locations in an image. I selected the small version of YOLOv8 because it provides a good balance between speed and accuracy.

Training

The model was trained on Google Colab using a Tesla T4 GPU. The dataset was prepared in YOLO format and connected using a dataset.yaml file. Training was performed for 20 epochs with an image size of 640 and a batch size of 16.

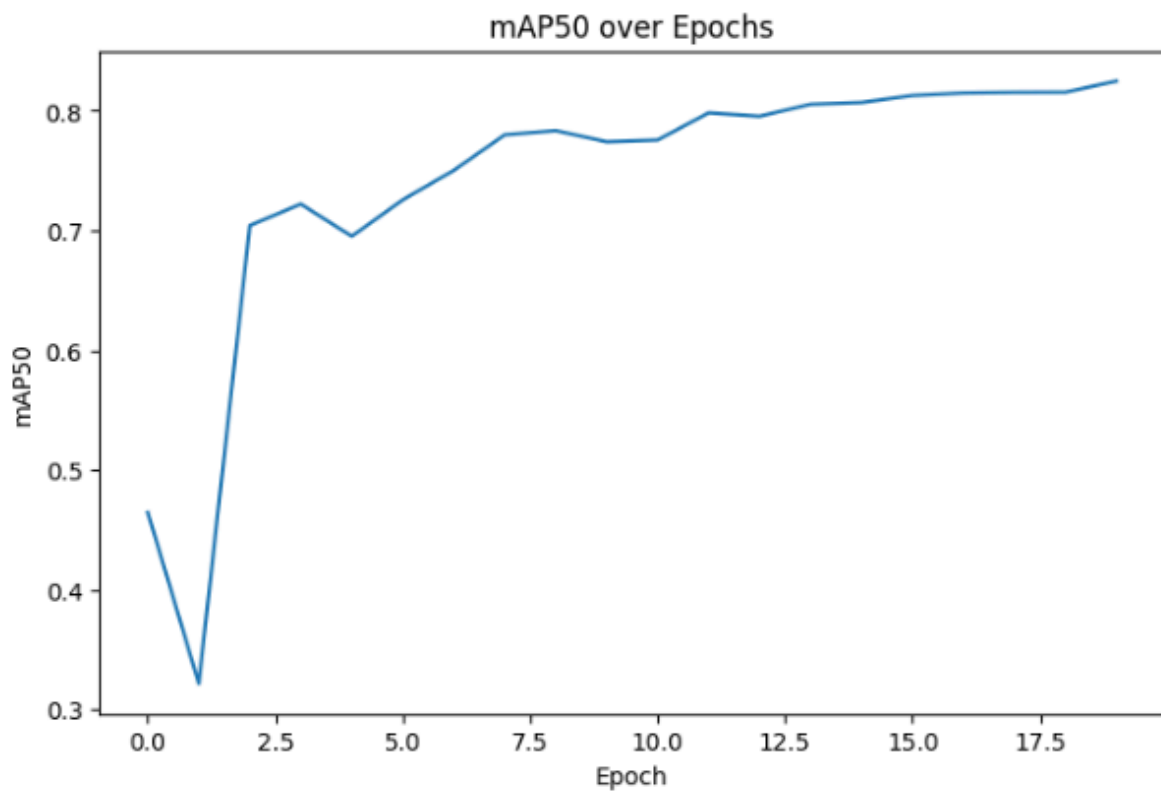
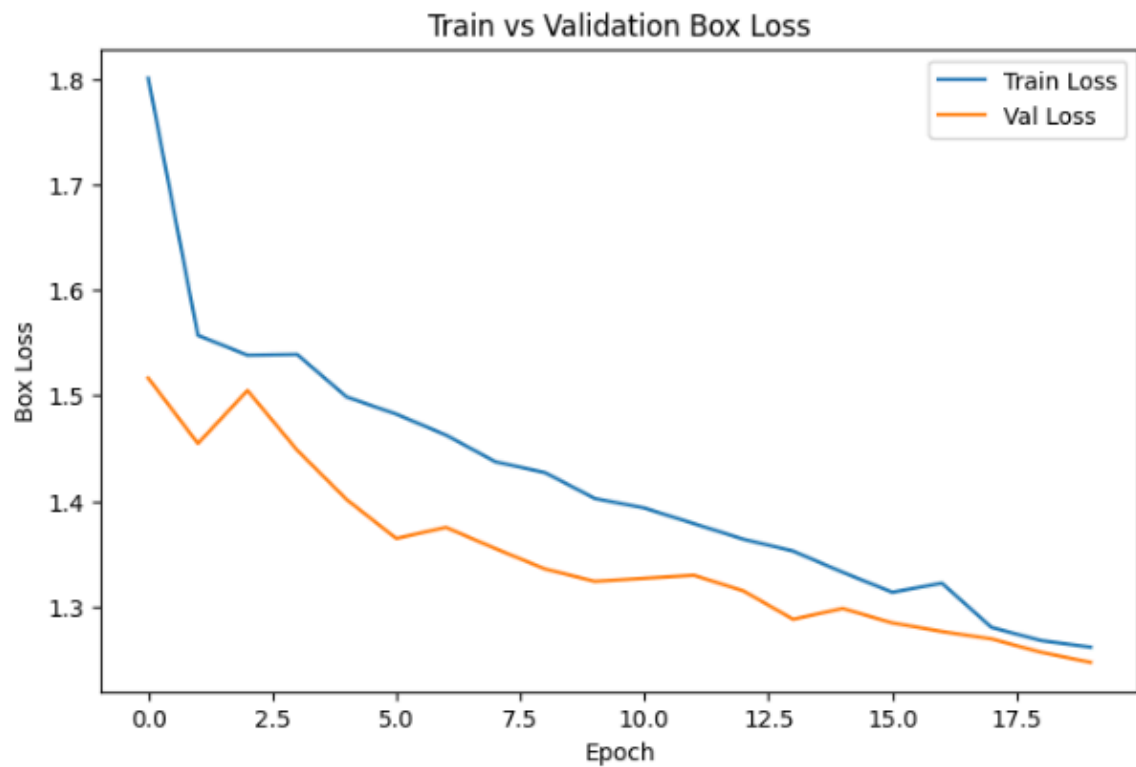
During training, both training loss and validation loss were monitored. The model performance improved over the epochs and the loss gradually decreased.

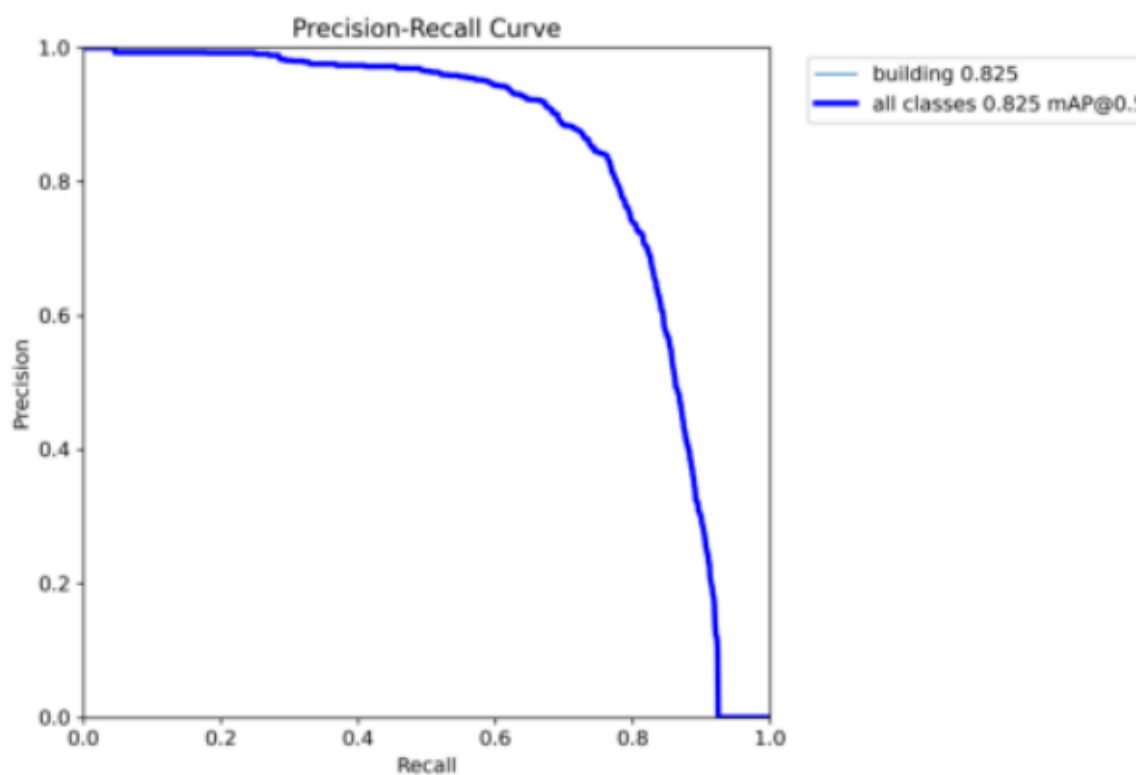
Results

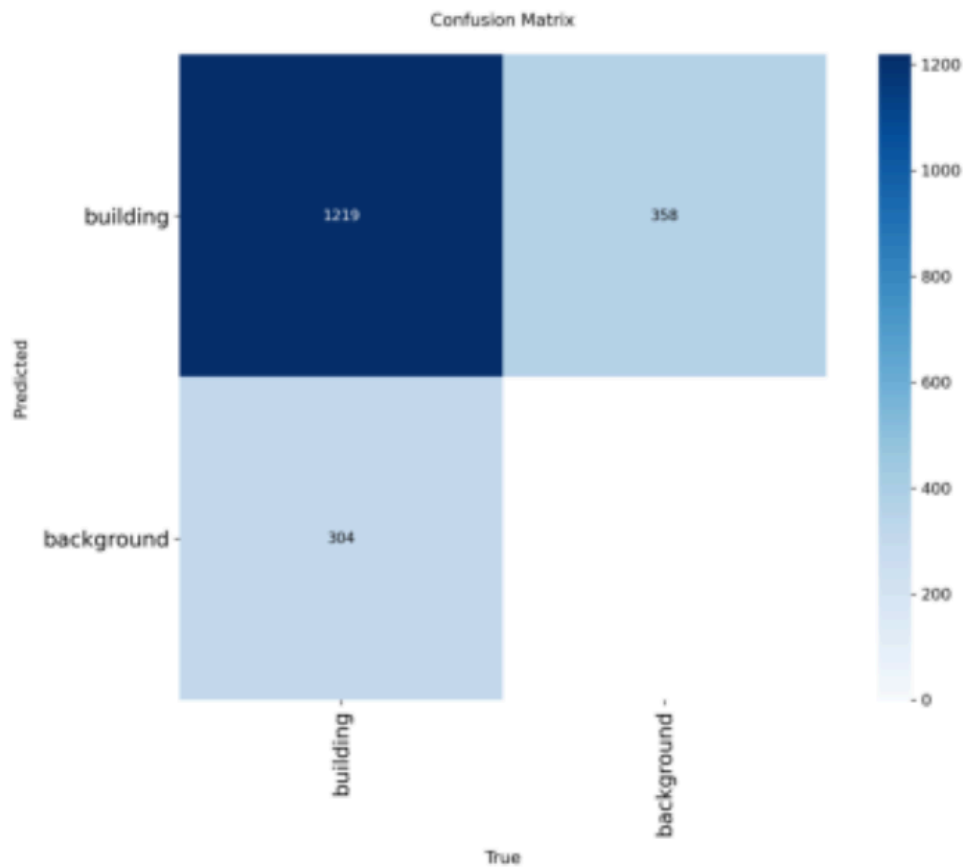
The trained model achieved the following results:

- Precision: 0.8445
- Recall: 0.7491
- mAP@50: 0.8245
- F1 Score: 0.794

The model was able to detect most buildings correctly and showed good overall performance on the test dataset.







Processing time: 38.9 ms

Original



Detected Buildings

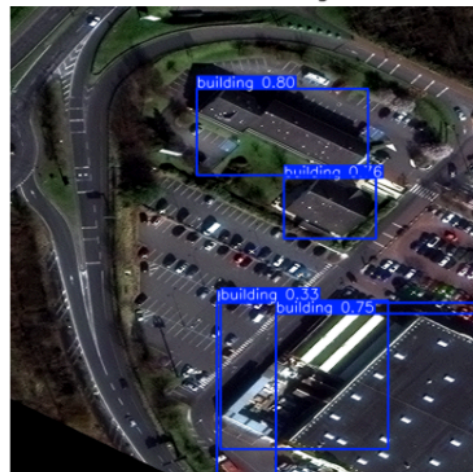


Image: AOI_3_Paris_img164.png
 Buildings detected: 5
 Avg confidence: 0.58
 Processing time: 36.3 ms

Metric	Your value	What it means
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mAP@50	0.8245	Overall detection accuracy of the model
Precision	0.8445	Percentage of predicted buildings that were correct
Recall	0.7491	Percentage of actual buildings detected by the model
F1 score	0.794	Balance between precision and recall

Observations

The model performed well on most images and was able to detect buildings with good accuracy. It worked especially well when buildings were clearly visible and separated from each other.

However, the model sometimes missed buildings in very dense areas where many buildings were located close together. Small rooftops and shadows also created some challenges.